

SANTHOSH S

PCB Design Enthusiast | Aspiring Hardware / Embedded Engineer

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in PCB design, hardware integration, testing, and troubleshooting. Skilled in taking electronic prototypes from schematic design and board layout through fabrication, assembly, and validation, with practical exposure to IoT-based embedded systems. Seeking an entry-level opportunity in PCB Design, Hardware Design, or Embedded Systems.

TECHNICAL SKILLS

- **PCB Design Tools:** KiCad, Altium Designer (Learning)
- **PCB Fabrication Tools:** FlatCAM, LinuxCNC
- **CAD & 3D Modeling:** Autodesk Fusion 360
- **Programming Languages:** Basic C
- **Core Areas:** Embedded Systems, IoT, PCB Design, Hardware Integration & Testing

ACADEMIC PROJECTS

Smart Irrigation System Using IoT

- Developed a working IoT-based smart irrigation prototype with automatic and manual control modes for efficient plant watering.
- Implemented real-time monitoring of soil moisture, temperature, and humidity, with data displayed on an OLED screen and mobile dashboard.
- Designed a custom 2-layer PCB using KiCad and fabricated it through CNC milling, followed by soldering, assembly, testing, and troubleshooting.

AI-Powered PCB Inspection & Automated Defect Segregation

- Developing a YOLO-based computer vision system for automated PCB component inspection.
- Implementing component presence and absence detection to identify missing components on assembled PCBs.
- Developing component placement verification to validate whether the correct component is positioned at its designated reference location.
- Currently under development, with ongoing work focused on automated component-level PCB inspection and defect identification.

ESP32-WROOM-32 Development Board– KiCad

- Designed the schematic and PCB layout for a customized ESP32-WROOM-32 development board using KiCad.
- Created the PCB design with appropriate footprints, routing, and board layout considerations.

CERTIFICATIONS

- PCB Design Using KiCad – Workshop, CIET
- PCB Design Using Altium Designer – Altium Education (Ongoing)

EDUCATION

B.E. Electronics and Communication Engineering	2023 – 2027
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Coimbatore Institute of Engineering and Technology, Coimbatore CGPA: 7.08	
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Higher Secondary Certificate (HSC)	2022 – 2023
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Devi Metric Higher Secondary School, Thiruvarur	
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Secondary School Leaving Certificate (SSLC)	2020 – 2021
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